

Asymptotic form for CPA in spheres with finite surface impedance

We use the expressions found in the paper: Localized plasmons in graphene-coated nanospheres ([https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414?](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414?casa_token=QBJAm_2eqLcAAAAA%3A1mAgKobdxS6Z4f265WwfP6XgDlexclJHYqBr6tU6ASgxSmc7Hb2Mx7c57sp8cConRpcejFzZ0VPMQ)

https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414?casa_token=QBJAm_2eqLcAAAAA%3A1mAgKobdxS6Z4f265WwfP6XgDlexclJHYqBr6tU6ASgxSmc7Hb2Mx7c57sp8cConRpcejFzZ0VPMQ), and we use asymptotic forms for the spherical Bessel/Hankel functions from <https://www.damtp.cam.ac.uk/user/tong/aqm/bessel.pdf> (<https://www.damtp.cam.ac.uk/user/tong/aqm/bessel.pdf>).

In particular, for large $a\omega/c$, we can expand the spherical Bessel functions as:

$$j_l(x) \approx \frac{\sin(x - l\pi/2)}{x}, y_l(x) \approx -\frac{\cos(x - l\pi/2)}{x} \rightarrow$$

$$h_l^{(1)}(x) \approx -i \frac{\exp(i(x - l\pi/2))}{x}, h_l^{(2)}(x) \approx i \frac{\exp(-i(x - l\pi/2))}{x}$$

Furthermore, we may write the derivatives of the Ricatti-Bessel functions as:

$$\left[x j_l(x) \right]' \approx \cos(x - l\pi/2), \left[x h_l^{(2)}(x) \right] \approx \exp(-i(x - l\pi/2)),$$

$$\left[x h_l^{(1)}(x) \right]' \approx \exp(i(x - l\pi/2))$$

Which means the numerator of the (**TM**) scattering coefficients may be written as:

$$-h_l^{(2)}(x_2) \left[x_1 j_l(x_1) \right]' + \varepsilon_m j_l(x_1) \left[x_2 h_l^{(2)}(x_2) \right]' + g(\omega) \left[x_1 j_l(x_1) \right]' \left[x_2 h_l^{(2)}(x_2) \right]',$$

where $\varepsilon_m \equiv n_m^2$ is the dielectric function of the sphere, $x_1 = n_m a\omega/c$, $x_2 = a\omega/c$ and $g(\omega) = \frac{i\sigma(\omega)}{\omega a \varepsilon_0}$, where $\sigma(\omega)$ is the conductivity of the surface layer (minus inverse of surface impedance). We may write the asymptotic expansion of the above as:

$$-\frac{i}{x_2} \exp(-i(x_2 - l\pi/2)) \cos(x_1 - l\pi/2) + \frac{\varepsilon_m}{x_1} \sin(x_1 - l\pi/2) \exp(-i(x_2 - l\pi/2))$$

$$+ g(\omega) \cos(x_1 - l\pi/2) \exp(-i(x_2 - l\pi/2)) = 0$$

For this to be zero, we require:

$$-i \cos(x_1 - l\pi/2) + n_m \sin(x_1 - l\pi/2) + \frac{i\sigma(\omega)}{c\epsilon_0} \cos(x_1 - l\pi/2) = 0$$

Note that if we have $\sigma(\omega) = c\epsilon_0$ (equivalently $g(\omega) = ic/\omega a = i/x_2$) (one half of the conductivity required for CPA for a monolayer thin film), we will have, for any frequency satisfying $n_m \omega a/c = l\pi/2 + m\pi$ (where m is any integer), that the scattered field will vanish and we will have CPA. Note that this also necessarily implies that modes with the same parity of l will all be absorbed at the same frequencies. Note that in the absence of a finite surface impedance, we can only get CPA if the following condition is met:

$$n_m \approx i \cot(x_1 - l\pi/2)$$

We also examine the denominator of the TM scattering coefficients to make sure they do not also vanish. This denominator is given by:

$$h_l^{(1)}(x_2) [x_1 j_l(x_1)]' - \epsilon_m j_l(x_1) [x_2 h_l^{(1)}(x_2)]' - g(\omega) [x_1 j_l(x_1)]' [x_2 h_l^{(1)}(x_2)]',$$

Note that in the small $a\omega/c$ limit, the expansion of this term gives the quasistatic plasmon dispersion. We expand this in the large $a\omega/c$ limit, as we did with the numerator, and obtain (up to an overall factor of $\exp(i(x_2 - l\pi/2))$):

$$-\frac{i}{x_2} \cos(x_1 - l\pi/2) - \frac{\epsilon_m}{x_1} \sin(x_1 - l\pi/2) - g(\omega) \cos(x_1 - l\pi/2)$$

For this to be zero, we need:

$$i \left[\sigma/c\epsilon_0 + 1 \right] \cos(x_1 - l\pi/2) + n_m \sin(x_1 - l\pi/2) = 0,$$

which clearly cannot be satisfied at the CPA frequencies derived above.

Asymptotic expansion for the CPA condition for TE modes

We now do the same expansion for the TE modes:

The numerator for the TE modes is given by:

$$-j_l(x_1) \left[x_2 h_l^{(2)}(x_2) \right]' + h_l^{(2)}(x_2) \left[x_1 j_l(x_1) \right]' - g(\omega) x_0^2 j_l(x_1) h_l^{(2)}(x_2)$$

Asymptotically, this gives us (up to an overall factor of $\exp(-i(x - l\pi/2))$):

$$-\frac{\sin(x_1 - l\pi/2)}{x_1} + \frac{i}{x_2} \cos(x_1 - l\pi/2) - ig(\omega) x_0^2 \frac{\sin(x_1 - l\pi/2)}{x_1 x_2}$$

For this to vanish (for a real valued conductivity), then, we require

$$1 + ig(\omega) x_0^2 / x_2 = 0 \rightarrow 1 - \frac{\sigma(\omega) c}{a^2 \omega^2 \epsilon_0} \frac{a^2 \omega^2}{c^2} = 0$$

as well as $n_m \omega / c = l\pi/2 + (2m + 1)(\pi/2)$

Therefore, we require $g(\omega) = i/x_2$ (equivalently $\sigma(\omega) = c\epsilon_0$).

Can we absorb both TE and TM modes? To absorb all TM modes of a given parity we require $\sigma = c\epsilon_0$, which is the same condition as for the TE modes. Furthermore, we see that at frequencies at which odd TM modes are absorbed, we absorb all even TE modes, and vice-versa.